Problem 1

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Problem 1 *For each of the following functions, use the "short cut derivative rules" to compute their derivative.*

(a) $f(x) = \sqrt{x}$

Explanation.

$$f(x) = \sqrt{x} = x^{\frac{1}{2}}.$$

$$\begin{aligned} f'(x) &= \frac{1}{2}x^{\frac{1}{2}-1} \\ &= \frac{1}{2}x^{-\frac{1}{2}} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

(b) $s(u) = \dfrac{5}{u^2}$

Explanation.

$$s(u) = \frac{5}{u^2} = 5u^{-2}.$$

$$\begin{aligned} s'(u) &= 5(-2)u^{-2-1} \\ &= -10u^{-3} \\ &= \frac{-10}{u^3} \end{aligned}$$

(c) $p(t) = t^5 + 4t^3 + \pi$

Explanation.

$$\begin{aligned} p'(t) &= 5t^{5-1} + 4(3)t^{3-1} + 0 \\ &= 5t^4 + 12t^2 \end{aligned}$$

Note that $\dfrac{d}{dx}(\pi) = 0$ because π is a constant.